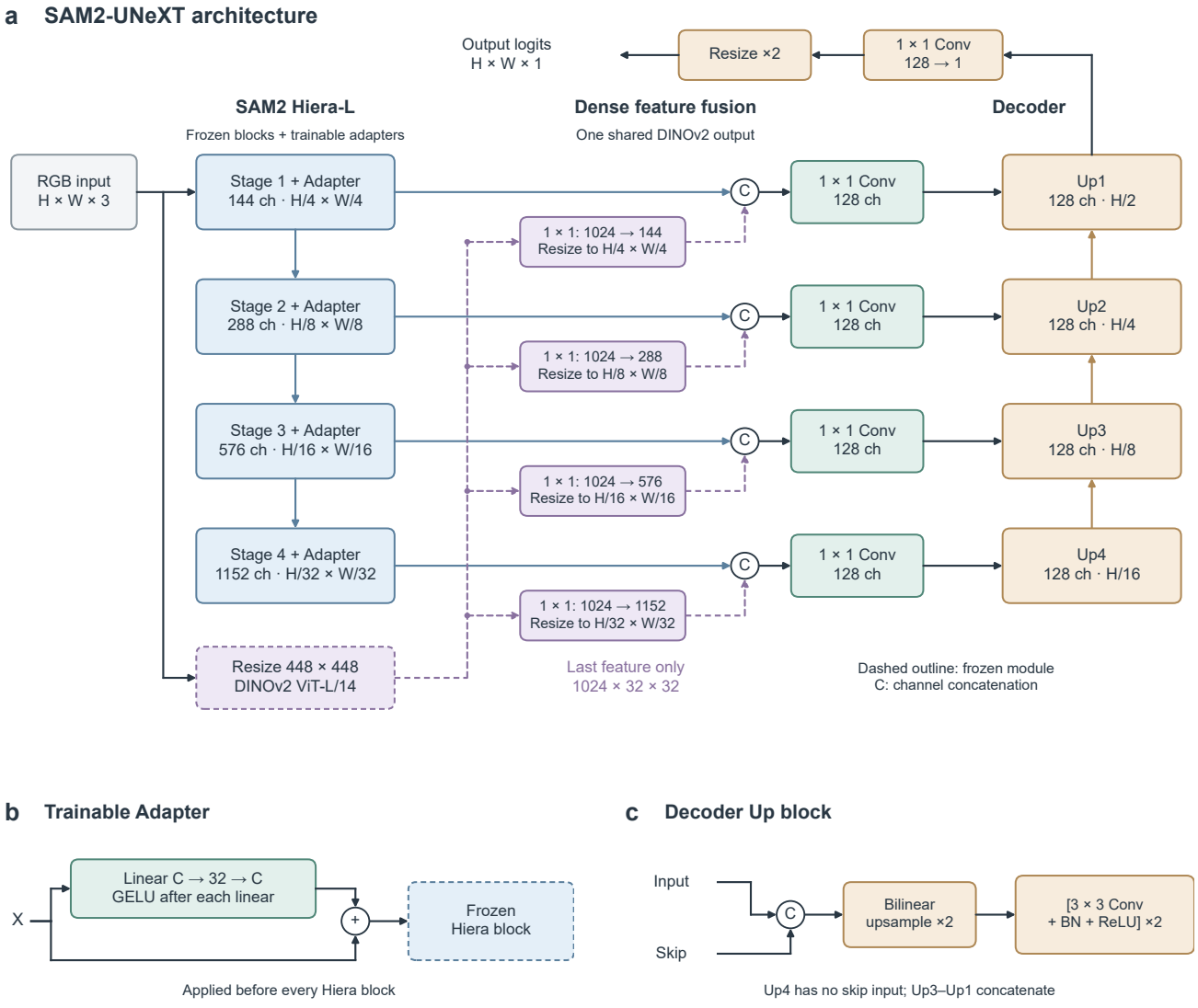

a SAM2-UNeXT architecture
Output logits
H × W × 1
Resize ×2
1 × 1 Conv
128 → 1
SAM2 Hiera-L
Frozen blocks + trainable adapters
Dense feature fusion
One shared DINOv2 output
Decoder
RGB input
H × W × 3
Stage 1 + Adapter
144 ch · H/4 × W/4
Stage 2 + Adapter
288 ch · H/8 × W/8
Stage 3 + Adapter
576 ch · H/16 × W/16
Stage 4 + Adapter
1152 ch · H/32 × W/32
Resize 448 × 448
DINOv2 ViT-L/14
1 × 1: 1024 → 144
Resize to H/4 × W/4
1 × 1: 1024 → 288
Resize to H/8 × W/8
1 × 1: 1024 → 576
Resize to H/16 × W/16
1 × 1: 1024 → 1152
Resize to H/32 × W/32
C
1 × 1 Conv
128 ch
Up1
128 ch · H/2
Up2
128 ch · H/4
Up3
128 ch · H/8
Up4
128 ch · H/16
Last feature only
1024 × 32 × 32
Dashed outline: frozen module
C: channel concatenation
b Trainable Adapter
X
Linear C → 32 → C
GELU after each linear
+
Frozen
Hiera block
Applied before every Hiera block
c Decoder Up block
Input
Skip
C
Bilinear
upsample ×2
[3 × 3 Conv
+ BN + ReLU] ×2
Up4 has no skip input; Up3–Up1 concatenate